

CVPR'26 Workshop: BigMAC: Big Model Adaptation for Computer Vision

BigMAC Organizing Team

3 November, 2025

1 Topic

Abstract. This workshop explores how large pretrained models are revolutionizing computer vision. We examine emerging techniques where models like Stable Diffusion enable image-to-3D reconstruction, architectures like VGG Transformer (VGGT) leverage pretrained DINO for enhanced visual understanding, and methods like REA harness pretrained models for improved diffusion generation. The workshop covers four key areas: emerging uses of pretrained vision models for 3D and generative AI, prompting techniques for vision models, the role of vision models in multimodal LLMs, and post-pretraining adaptation strategies. BigMAC brings together researchers to address the challenges and opportunities in effectively adapting foundation models for diverse downstream vision tasks.

Workshop title	BigMAC: Big Model Adaptation for Computer Vision
Workshop acronym	BigMAC
Workshop track	1) Foundation models, 2) Emerging topics
Edition	2nd
Keywords	Big Model Adaptation, Vision Foundation Models
Half or full day	half day
Anticipated audience size	300
Requested number of poster boards	30
Papers published in proceedings?	No
Special requests	No

Topics:

- Emerging uses of pretrained vision models for 3D and generative AI
- Prompting methods and techniques for pretrained vision models
- Role of vision models in Multi- and omnimodal LLMs
- Post-pretraining of vision foundation models
- Cross-domain transfer and robustness of adapted vision models

Broader impact statement. The goal of this workshop is to explore and discuss ways of effectively adapting and utilizing large pretrained models in computer vision. The sheer parameter and training dataset sizes mean that these foundation models often cannot be trained from scratch by academia, yet they offer unprecedented opportunities for downstream adaptation. These developments bring both challenges and novel opportunities - from leveraging pretrained diffusion models for 3D reconstruction to incorporating vision encoders into multimodal systems. Rather than viewing model scale

as a barrier, we focus on innovative usage and adaptation techniques and post-pretraining strategies that make these powerful models accessible and useful for diverse applications. In this workshop, we aim to bring together researchers from academia and industry to discuss how the vision community can best harness these pretrained foundations for advancing computer vision research and applications.

Ethical considerations around the topic. BigMAC touches upon ethics in AI: large-scale pre-trained models that demonstrate strong performance are increasingly being deployed in real-world applications. As these foundation models become central to computer vision systems, understanding their limitations and biases becomes critical. Given the growing importance of these models in shaping computer vision applications, we invite researchers from both academia and industry to share diverse perspectives and insights. This diversity of viewpoints is essential for developing responsible adaptation practices, identifying potential pitfalls early, and ensuring that the benefits of these powerful pretrained models are accessible while mitigating their risks. Through bringing together varied experiences and backgrounds, we aim to foster critical discussions about the ethical implications of widespread model adaptation and deployment in computer vision. A key to this will be our planned panel discussion.

Relationship to previous workshops:

- **Foundation Models.** Workshop on Foundation Models (CVPR 2023/2024), What is Next in Multimodal Foundation Models? (ICCV 2023/2025, CVPR 2024/2025), Emergent Visual Abilities and Limits of Foundation Models (ECCV 2024, CVPR 2025), Foundation Models for Medical Vision (CVPR 2024), Workshop on Large Foundation Models for Educational Assessment (NeurIPS 2024) → In addition to surveying general progress or domain-specific applications such as medical or education, we focus on concrete adaptation strategies. We emphasize post-pretraining, prompting, fine-tuning, and cross-modal integration rather than model scaling or training from scratch.
- **Efficient learning.** Efficient Deep Learning for Computer Vision (CVPR 2023), Workshop on Resource Efficient Deep Learning for Computer Vision (ICCV 2023), Efficient Large Vision Models (CVPR 2024), Workshop on Efficient Deep Learning for Computer Vision (CVPR 2024), Efficient Deep Learning for Foundation Models (ECCV 2024) → In addition to efficient deep learning, we welcome papers with a broader scope, that extends to data, labels, model parameters, and computational efficiency as constraints.
- **Un-/self-supervised learning.** SSL-WIN (ECCV 2022/2024), Self-supervised Learning - Theory and Practice (NeurIPS 2022/2023/2024) → In addition to papers presented in the workshops, we will also discuss the fundamental topic in order to solve dataset-related issues. Yuki M. Asano has been the primary organizer in both workshops, therefore, we will consider and arrange the similar topics.

2 Organizers and speakers

List of organizers:

1. **Yuki M. Asano (Primary Organizer)**, Professor at the University of Technology Nuremberg (UTN) and head of the Fundamental AI (FunAI) Research Lab.
2. **Anna Kukleva**, Postdoctoral Researcher at Max-Planck Institute for Informatics in Saarbrücken
3. **Hirokatsu Kataoka**, Chief Senior Researcher at AIST and Academic Visitor at Visual Geometry Group, University of Oxford.

4. **Ryousuke Yamada**, Researcher at AIST and visiting postdoctoral researcher at the University of Technology Nuremberg.
5. **Yoshihiro Fukuhara**, Joint appointed research fellow at AIST and research fellow at Waseda University.
6. **Go Irie**, Professor at Tokyo University of Science.
7. **Ivona Najdenkoska**, Postdoctoral Researcher at University of Amsterdam
8. **Aida Nematzadeh**, Senior Staff Research Scientist at DeepMind
9. **Serge Belongie**, Professor at University of Copenhagen, head of Pioneer Centre for Artificial Intelligence

Primary and secondary organizers:

1. Primary contact name and email: Yuki Asano / yuki.asano@utn.de
2. Secondary contact name and email: Anna Kukleva / akukleva@mpi-inf.mpg.de

Organizers' experience and background:

- Yuki Asano: organized 9 workshops at top conferences: ICCV'25, NeurIPS'24, ECCV'24, CVPR'24, NeurIPS'23, ICCV'23, ECCV'22, NeurIPS'22, ECCV'20. Four of these as **main-organizer**, including the previous version of the BigMAC workshop at ICCV'23. He was also the main-organizer for two ELLIS PhD schools on Foundation Models in Amsterdam with >200 attendees.
- Anna Kukleva: organized four workshops in different roles at top conferences: ECCV'20, CVPR'24-25, NeurIPS'25. She also served as main organizer for Doctoral Consortium at ICCV'25.
- Hirokatsu Kataoka: has organized 6 workshops at top-tier conferences including ICCV'19, CVPR'21, ICCV'23, CVPR'24, ICCV'25, ICCV'25. Three of these as primary organizer and one for primary advisor.
- Ryousuke Yamada: organized three workshops at top conferences: ICCV'23, CVPR'24, and ICCV'25. He served as Program Chair and Publication Chair at these workshops. He also co-manages a large computer vision community in Japan.
- Yoshihiro Fukuhara: organized two workshops at ICCV 2025, serving as the main organizer for one of them. He has been managing a research community with 1,700 members in Japan focused on computer vision for more than ten years.
- Go Irie: organized four international workshops (ICME'13, IW-FCV'23-'25) and served on organizing committees for six major conferences including ACM Multimedia'12 and ACM ICMR'18/'22.
- Ivona Najdenkoska: main organizer for a planned ICLR'26 workshop, co-led research collaborations both in academia and industry, served as a reviewer for CVPR/ICCV/ECCV/ICLR/NeurIPS.
- Aida Nematzadeh: Organized workshops ECCV'24, ACL'22, NeurIPS'21 and ICLR'21.
- Serge Belongie: Organized more than ten editions of the FGVC workshop at CVPR and four editions of the CV4ARVR workshop at CVPR and ICCV. He was also Program Chair for CVPR 2012, a General Chair for WACV 2014, and will serve as a General Chair for ECCV 2026.

2.1 List of invited speakers

- **Oriane Simeoni** (Researcher at Meta FAIR)
 - (a) Confirmed
 - (b) Homepage: <https://osimeoni.github.io/>
 - (c) Relevance to the workshop: She works on the DINO team at Meta FAIR, advancing self-supervised vision encoders such as DINOv3 that transfer broadly with little or no annotation, which is a cornerstone for adapting large models to new tasks and domains. Adjacent post-training directions, for example, using DINO features to improve CLIP for zero-shot dense segmentation or aligning DINO features with text, exemplify light-weight and high-impact adaptation strategies central to the workshop.
- **Christian Rupprecht** (Professor at University of Oxford)
 - (a) Confirmed
 - (b) Homepage: <https://chrirupp.github.io/>
 - (c) Relevance to the workshop: His recent work, VGGT (CVPR’25 Best Paper Award), demonstrates a single feed-forward transformer that directly infers camera parameters, depth, point maps, and 3D tracks, and it then serves as a strong visual backbone for multiple 3D tasks—an archetype of “pretrain once, adapt broadly.” Complementary works such as CoTracker (joint point tracking) and RealFusion (using diffusion priors for single-image 3D reconstruction) illustrate practical pathways to repurpose powerful pre-trained models across video and geometric settings.
- **Cordelia Schmid** (Research director/Professor at INRIA, Researcher at Google)
 - (a) Confirmed
 - (b) Homepage: <https://thoth.inrialpes.fr/~schmid/>
 - (c) Relevance to the workshop: Her line of work on video understanding and grounding (e.g., Grounded Video Caption Generation) unifies captioning with temporally consistent object grounding and introduces large-scale automatic annotation, pushing scalable adaptation in video. Her perspective on aligning language with visual evidence in long videos maps directly to adapting foundation models for robust video reasoning and retrieval.
- **Saining Xie** (Professor at NYU Courant)
 - (a) Confirmed
 - (b) Homepage: <https://www.sainingxie.com/>
 - (c) Relevance to the workshop: His REPA and RAE show how strong vision encoders can be reused to stabilize and dramatically accelerate the training of large-scale diffusion transformers, achieving state-of-the-art quality with significantly less computation. This “representation-first” approach provides a compelling recipe for post-training or adapter-style adaptation of generative vision models. In addition, his recent works push toward spatial visual intelligence, a new frontier for large-scale pre-trained models that aligns directly with this workshop’s objectives.
- **Yannis Kalantidis** (Senior Researcher at NAVER LABS Europe)
 - (a) Confirmed
 - (b) Homepage: <https://www.skamalas.com/>

- (c) Relevance to the workshop: Relevance to the workshop: His UNIC and DUNE show how multi-teacher distillation can fuse diverse teacher models (including heterogeneous image and 3D teachers) into a single universal student model, improving transfer while reducing both serving and training costs. This directly advances the workshop’s goal of efficient adaptation by transforming many specialist teacher models into a single broadly capable student model.

- **Lucas Beyer** (Researcher at Meta Superintelligence Labs)

- (a) Confirmed
- (b) Homepage: <https://lucasb.eyer.be/>
- (c) Relevance to the workshop: He has been at the forefront of foundational work in vision and multimodal models from his influential works on distillation, ViT, SigLIP, Gemma and LiT. His research fits squarely in our big model adaptation theme.

2.2 Diversity

We have strived to achieve excellence in both speaker and organizer line-up whilst maintaining strong diversity. The gender ratio amongst our speakers is 4/2, among organisers 6/3. The industry/academia ratio is 3/3 among speakers. Our speakers and organizers come together from 12 institutions and 9 countries. Finally, among our speakers and organizers, we have also have a good mixture of all career stages from more junior to senior established researchers. Lastly, we also strived to achieve diverse topic coverage such that multiple perspectives are covered. The speakers cover varying backgrounds, including self-supervised learning, multimodal LLMs, Spatial AI, generative modeling and video understanding – all under the umbrella of Big Model Adaptation.

3 Format and logistics

Format: in-person

Expected workshop size: Large (300 attendees)

Request for full-day workshop: No

Schedule: The workshop will contain 6 invited talks, 1 panel sessions, and a 30 invited-only poster session. We are able to organize the workshop on either date during the CVPR 2026 workshop days. We will flexibly arrange the time schedule along with the invited speakers.

Time	Duration	Event
9:00 - 9:10	10 min	Opening remarks
9:10 - 9:30	20 min	Invited talk (1)
9:30 - 9:50	20 min	Invited talk (2)
9:50 - 10:10	20 min	Invited talk (3)
10:10 - 10:20	10 min	Short break
10:20 - 10:40	20 min	Invited talk (4)
10:40 - 11:00	20 min	Invited talk (5)
11:00 - 11:20	20 min	Invited talk (6)
11:20 - 11:30	10 min	Short break
11:30 - 12:15	45 min	Panel discussion (speakers 1-6)
12:15 - 13:30	75 min	Lunch break / poster session (invite-only)

Table 1: Workshop Schedule

For the lunch break, we will order vegan and non-vegan BigMac burgers and provide participants with a light-hearted snack, following the motto of *a shared meal brings a shared experience*.

Paper submission. No.

Competition. N/a

In-person posters. We're planning to set 30 posters.

Special requests. N/a